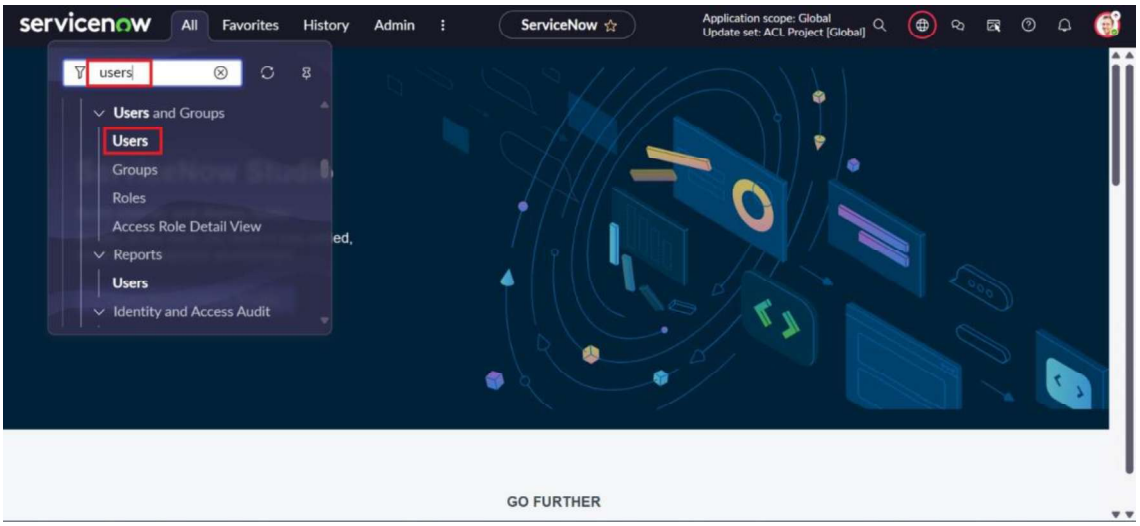
Script-Controlled ACL – Restrict Record Access Based on Field Value

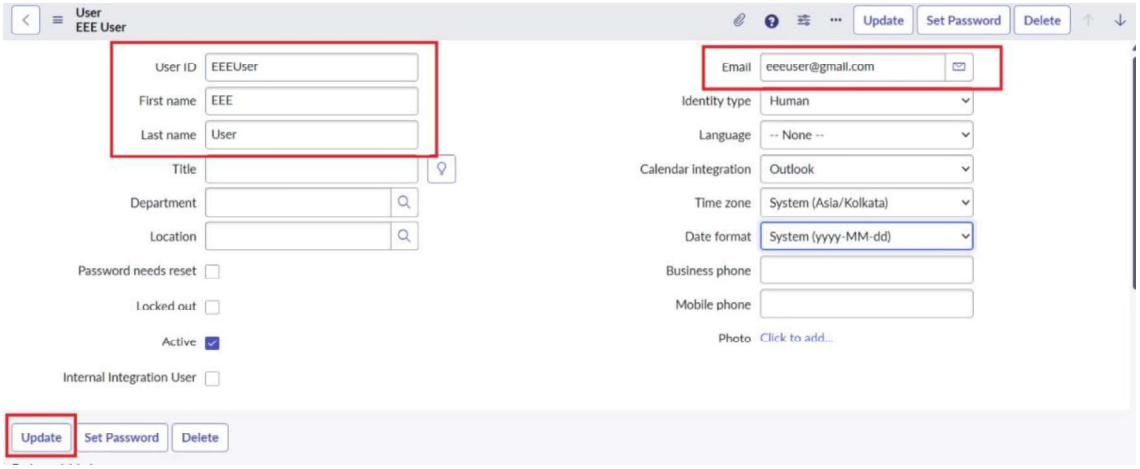
Users Creation:

Description:

1. Log in to the ServiceNow instance with admin access.
2. Create a test user:

- Navigate to User Administration → Users
- Click New
- User ID: EEE User
- First name: EEE
- Last name: User
- Email: eeeuser@gmail.com
- Save the record

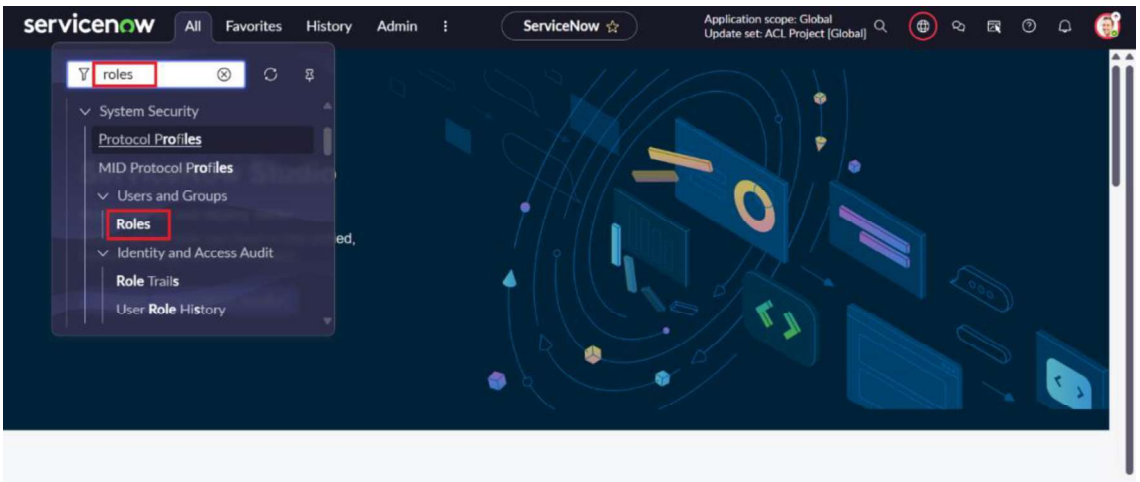


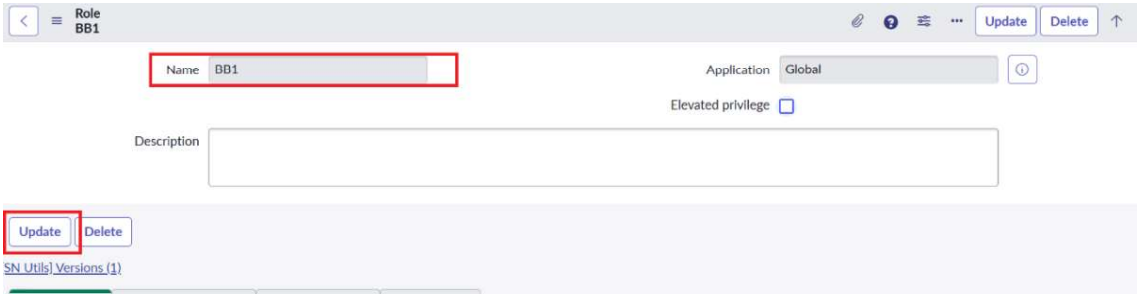


Roles Creation:

Description:

- Navigate to User Administration → Roles
- Click New
- Name: Enter role name as bb1
- Description: Brief role purpose
- Click Submit
- Create other roles also like bb2,bb3,bb4.(follow the same procedure)
- Assign all the roles to the EEE User

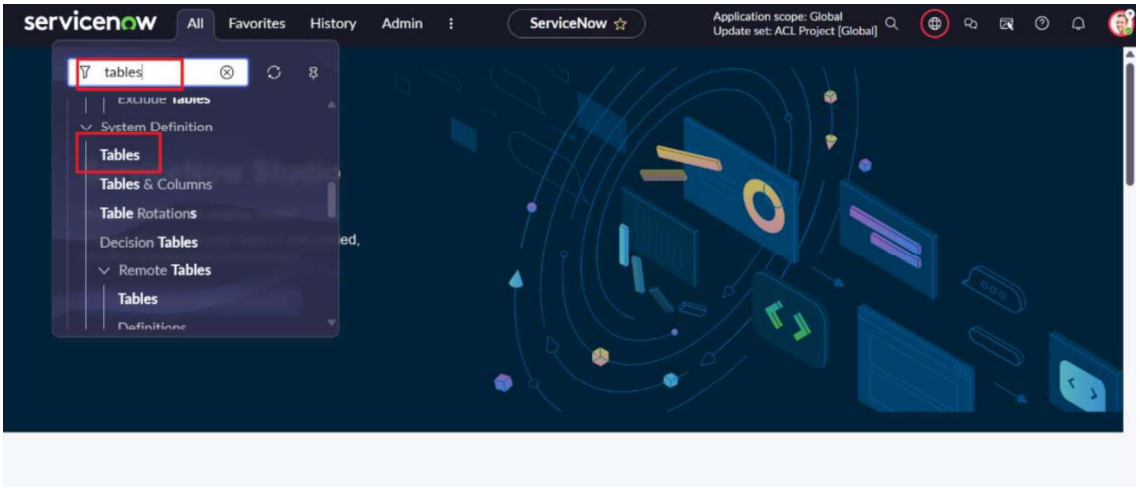


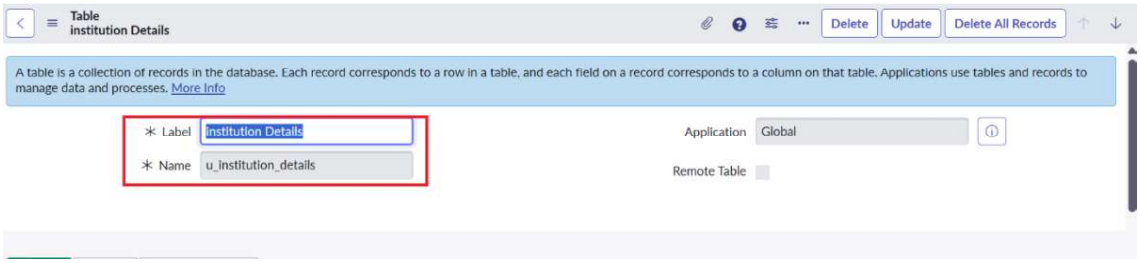


Creation of Table:

Description:

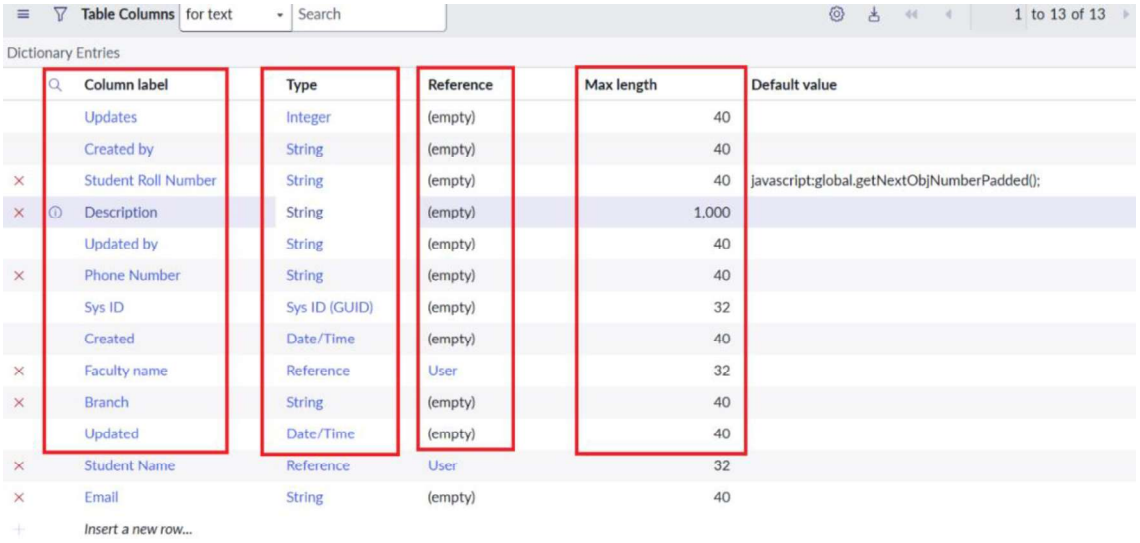
- In the Application Navigator, search for System Definition → Tables.
- Click New and create a custom table with the following details
- Label: Institution Details
- Name: u_institution_details
- Extends for : false
- Save or Submit the form



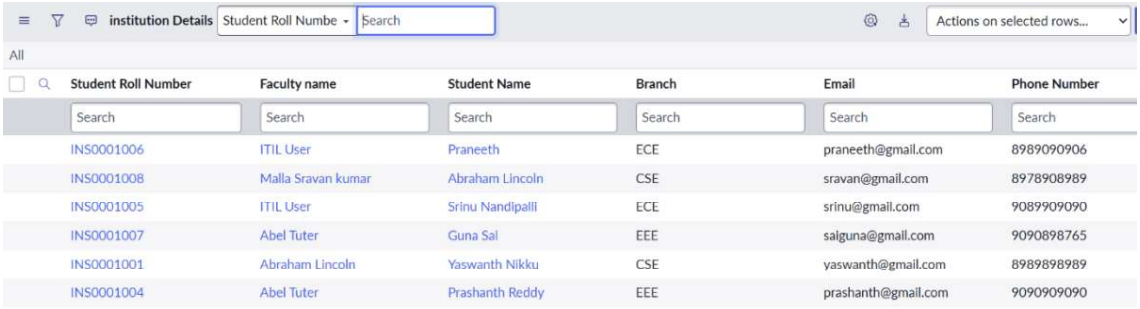


- In the columns tab, user Insert a New row method to create the mentioned fields as shown below picture
- The following fields to the u_institution_details table

- Student Roll Number– Auto Number
- Student Name – Reference- User
- Faculty Name – Reference- User
- Branch – Choice (ECE, EEE, CSE)
- Email– String
- Phone Number– String
- Description– Multi String
- Save the Table



- Click on the Show List in the Related Links
- Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).

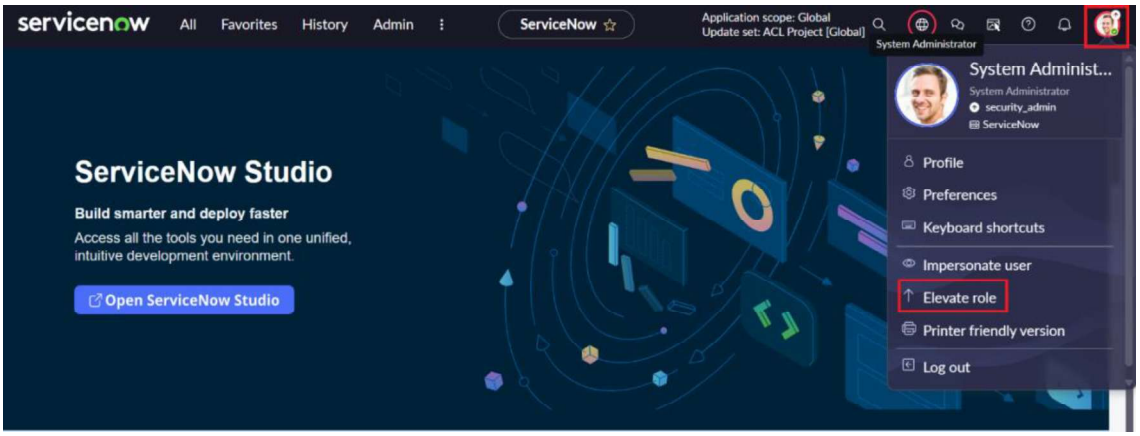


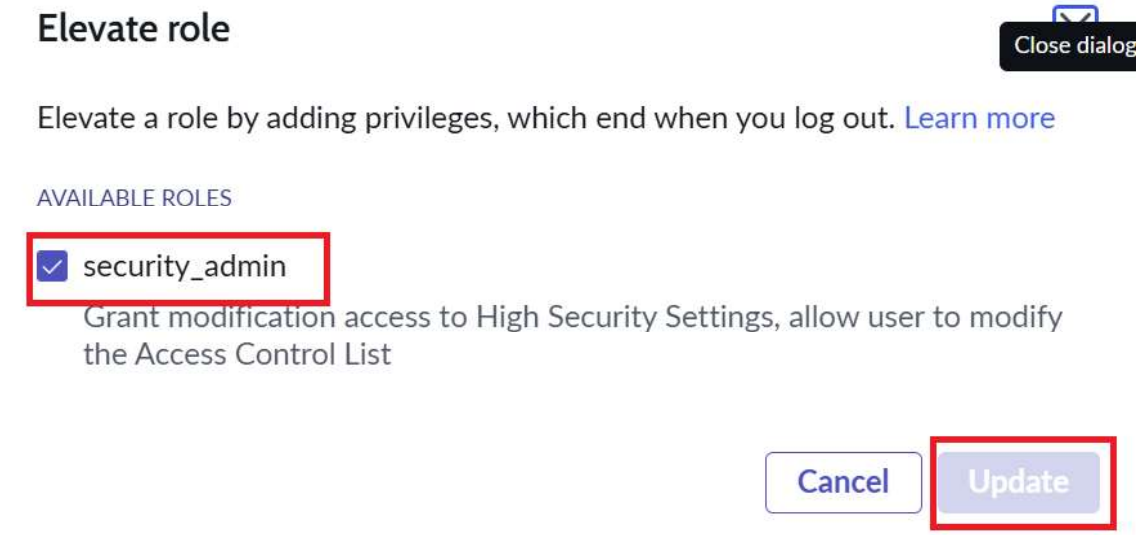
ACL Configuration:

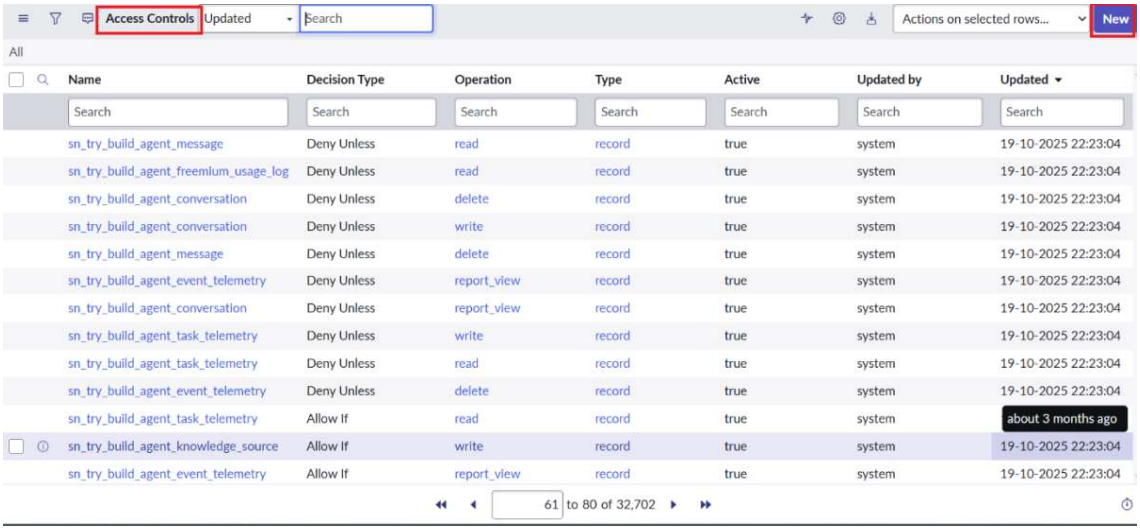
Description:

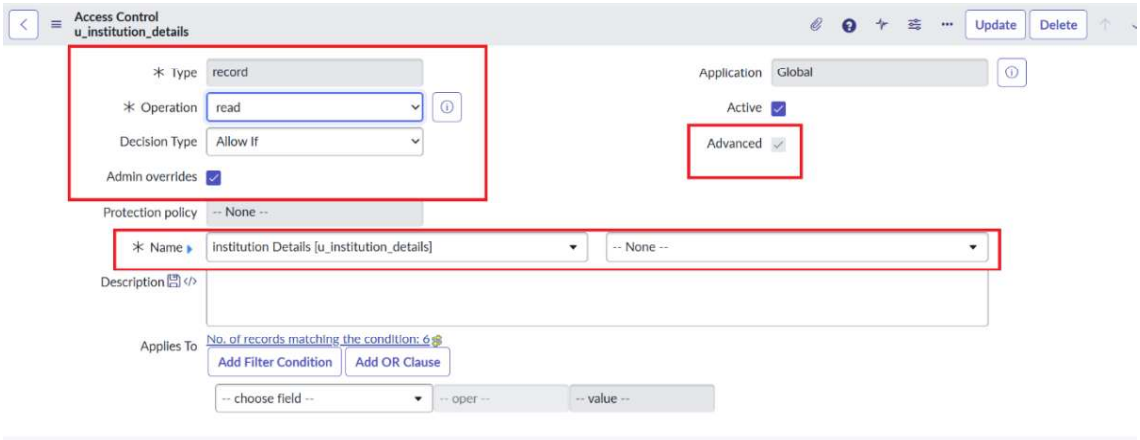
- In the Application Navigator, search for System Security → Access Control (ACL).
- Elevate “Security_admin” roll.
- Click New to create a new ACL.
- Enter the following details
- Type: record
- Operation: read
- Name: u_institution_details
- Active: true
- Advanced: true
- In the Requires role related list, add: bb1 (custom role)
- Add Data condition as Branch is EEE
- In the Script field, add the following script
- Save the ACL form

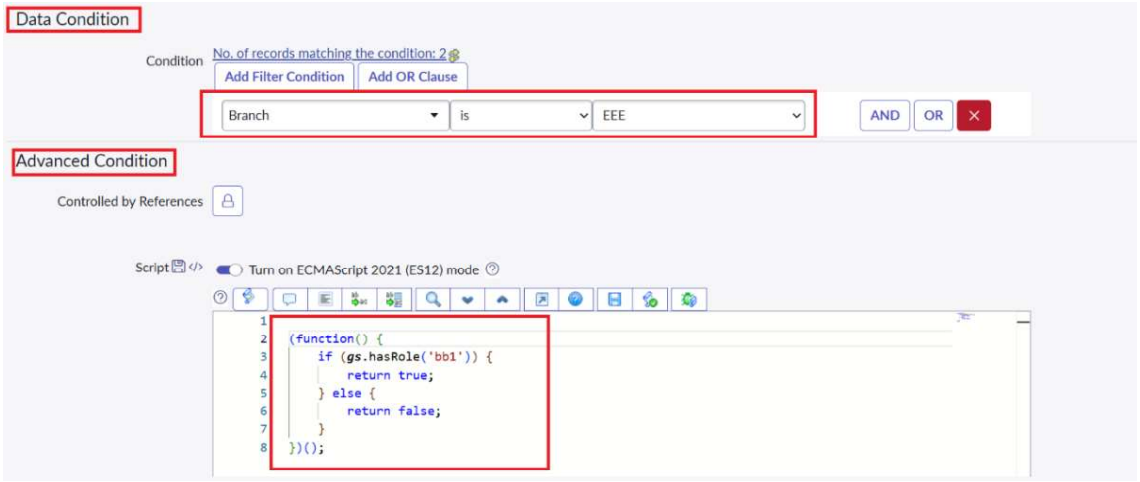
```
(function () {
   // Allow admin users full access
   if (gs.hasRole('admin')) {
      return true;
   }
   // Allow only EEE branch users to see EEE records
   if (gs.hasRole('bb1')){
      return true;
   }
   // Deny access for all others
   return false;
})();
```


servicenow
All
Favorites
History
Admin
ServiceNow
Application scope: Global
Update set: ACL Project [Global]
System Administrator
System Administ...
System Administrator
security_admin
ServiceNow
Profile
Preferences
Keyboard shortcuts
Impersonate user
Elevate role
Printer friendly version
Log out
ServiceNow Studio
Build smarter and deploy faster
Access all the tools you need in one unified, intuitive development environment.
Open ServiceNow Studio


Elevate role
Close dialog
Elevate a role by adding privileges, which end when you log out. Learn more
AVAILABLE ROLES
security_admin
Grant modification access to High Security Settings, allow user to modify the Access Control List
Cancel
Update

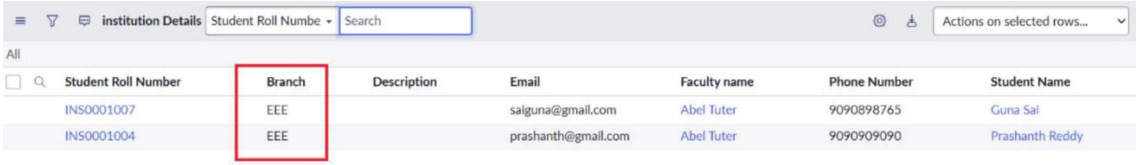

Access Controls
Updated
Search
Actions on selected rows...
New
All
Name
Decision Type
Operation
Type
Active
Updated by
Updated
sn_try_build_agent_message Deny Unless read record true system 19-10-2025 22:23:04
sn_try_build_agent_freemium_usage_log Deny Unless read record true system 19-10-2025 22:23:04
sn_try_build_agent_conversation Deny Unless delete record true system 19-10-2025 22:23:04
sn_try_build_agent_conversation Deny Unless write record true system 19-10-2025 22:23:04
sn_try_build_agent_message Deny Unless delete record true system 19-10-2025 22:23:04
sn_try_build_agent_event_telemetry Deny Unless report_view record true system 19-10-2025 22:23:04
sn_try_build_agent_conversation Deny Unless report_view record true system 19-10-2025 22:23:04
sn_try_build_agent_task_telemetry Deny Unless write record true system 19-10-2025 22:23:04
sn_try_build_agent_task_telemetry Deny Unless read record true system 19-10-2025 22:23:04
sn_try_build_agent_event_telemetry Deny Unless delete record true system 19-10-2025 22:23:04
sn_try_build_agent_task_telemetry Allow If read record true system about 3 months ago
sn_try_build_agent_knowledge_source Allow If write record true system 19-10-2025 22:23:04
sn_try_build_agent_event_telemetry Allow If report_view record true system 19-10-2025 22:23:04
61 to 80 of 32,702



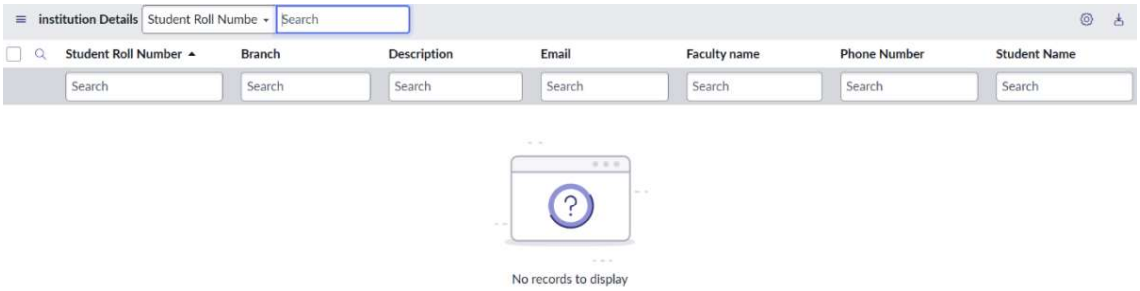


- Impersonate EEE users and open the Student Records list.
- Copy the table name u_institution_details.list/LIST and paste in the All menu.
- Impersonate Admin and open the Student Records list.

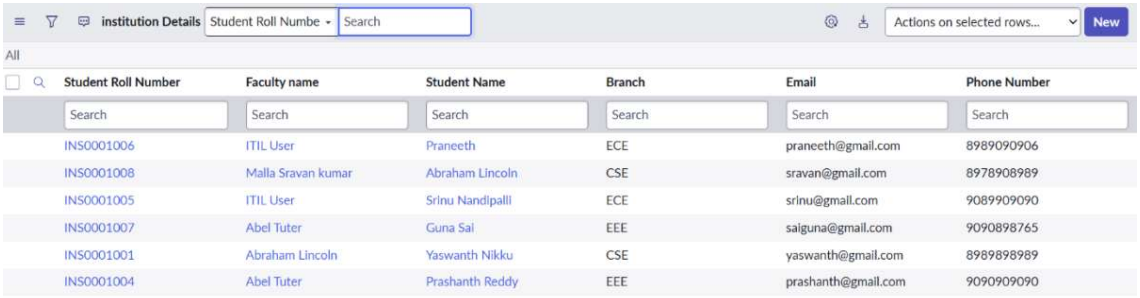
Verify : User with bb1 role can view only EEE branch records



- User without the role cannot view any records



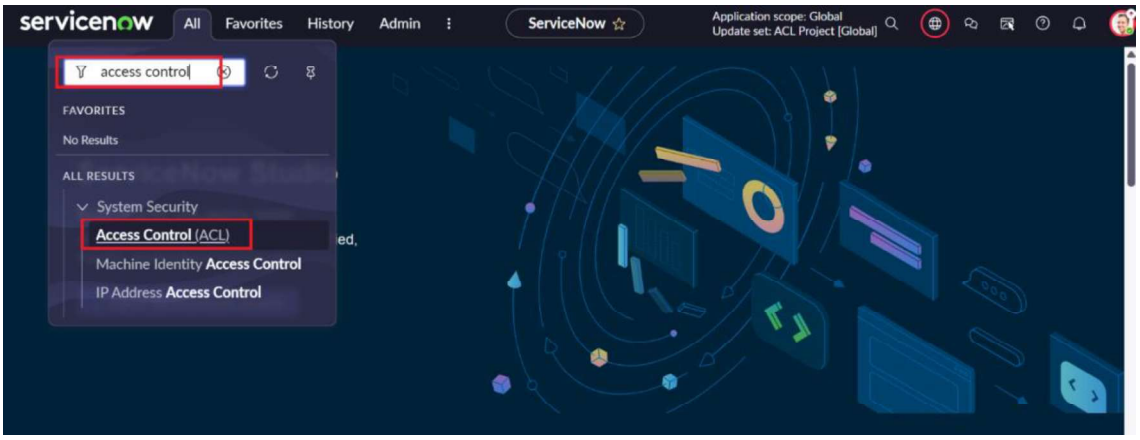
- Admin user can view **all records regardless of branch**

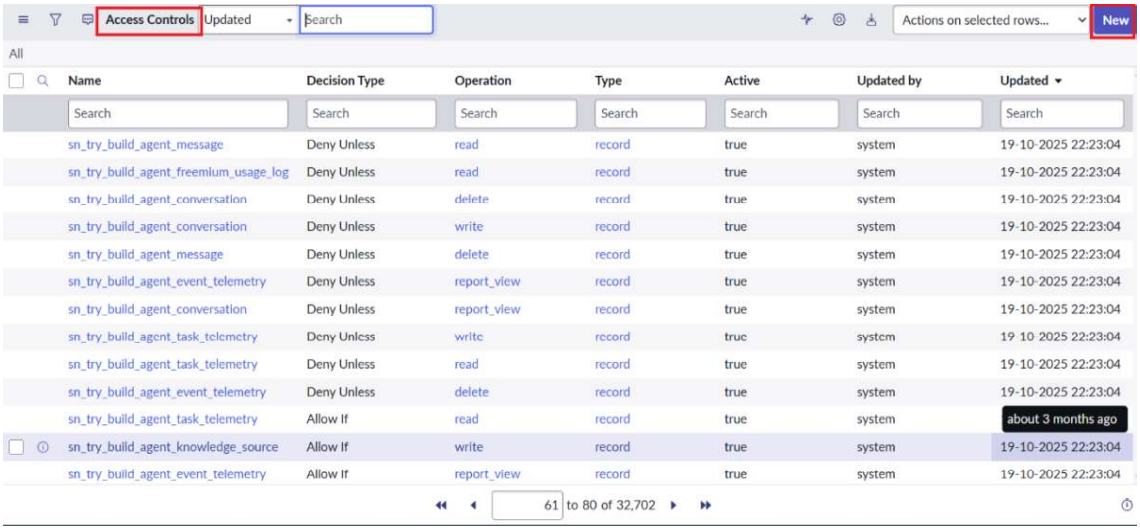


Creation of ACL-Create:

Description:

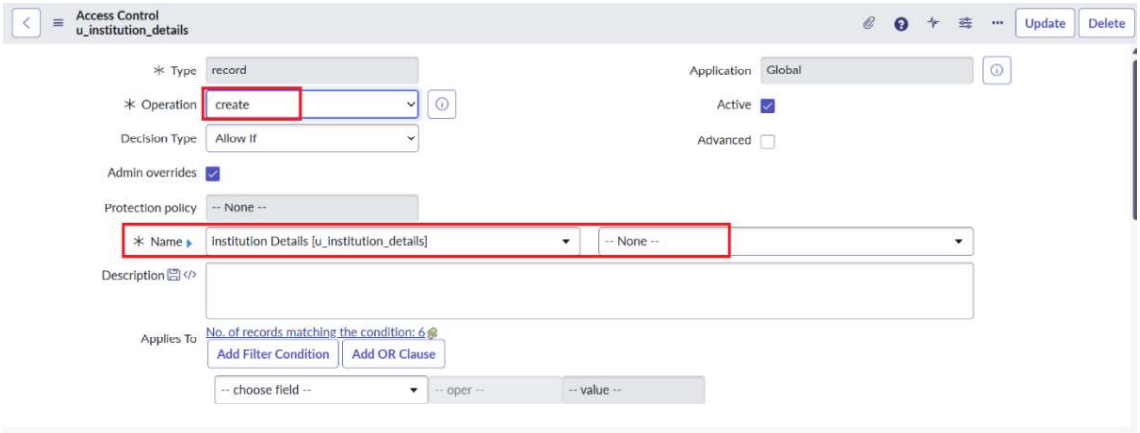
- Click New to create a new ACL(Create)
- Enter the following details
- Type: record
- Operation: create
- Name: u_institution_details
- Active: true
- In the Requires role related list, add: bb2 (custom role)
- No need to add any Data condition
- Save the ACL Record

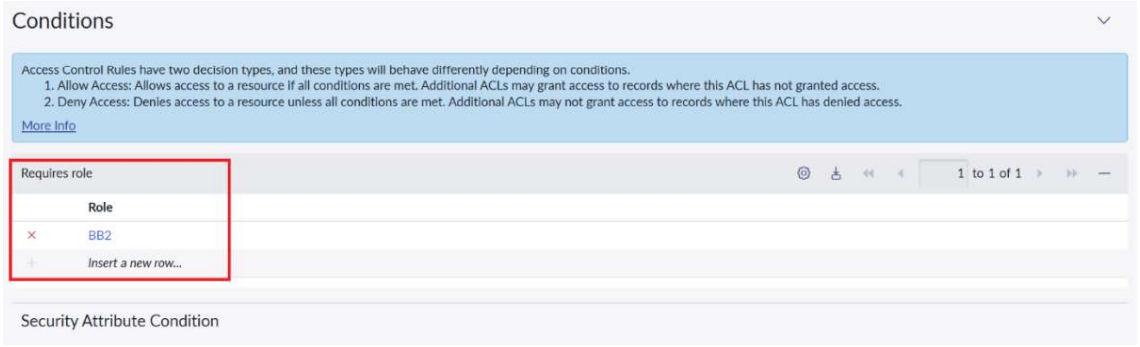




Name	Decision Type	Operation	Type	Active	Updated by	Updated
sn_try_build_agent_message	Deny Unless	read	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_freemium_usage_log	Deny Unless	read	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_conversation	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_conversation	Deny Unless	write	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_message	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_event_telemetry	Deny Unless	report_view	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_conversation	Deny Unless	report_view	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_task_telemetry	Deny Unless	write	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_task_telemetry	Deny Unless	read	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_event_telemetry	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_task_telemetry	Allow If	read	record	true	system	about 3 months ago
sn_try_build_agent_knowledge_source	Allow If	write	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_event_telemetry	Allow If	report_view	record	true	system	19-10-2025 22:23:04

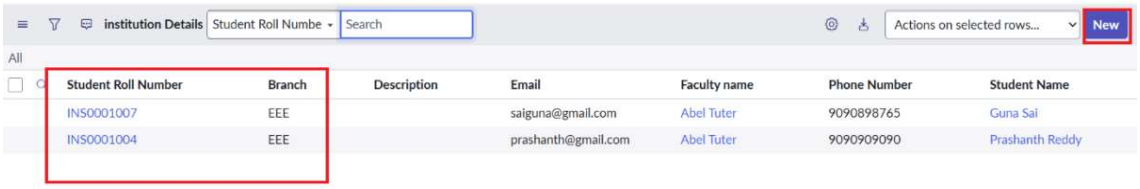
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Verify :

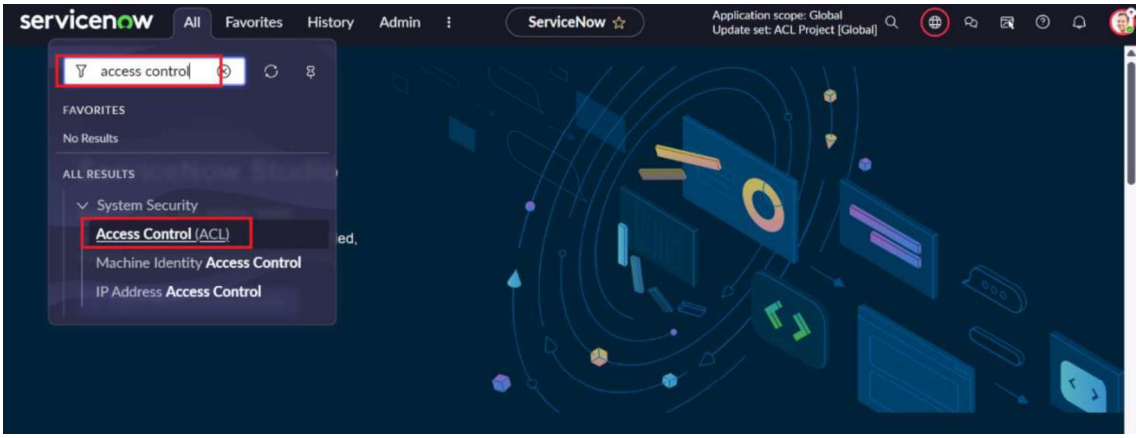
- User with **bb1 &bb2 roles** can view **only EEE branch records & New** Button

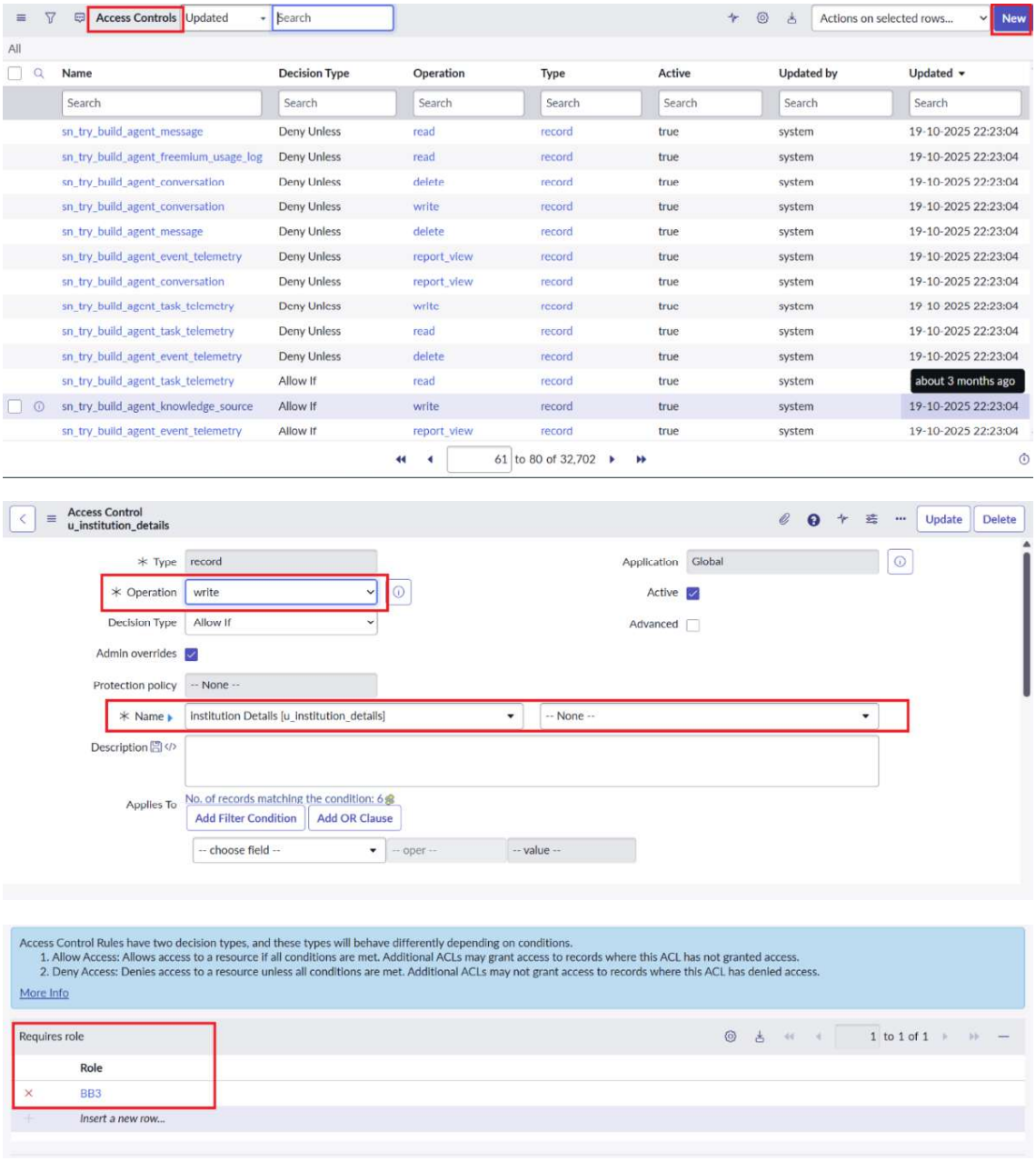


<u>Creation of ACL-Write:</u>

Description:

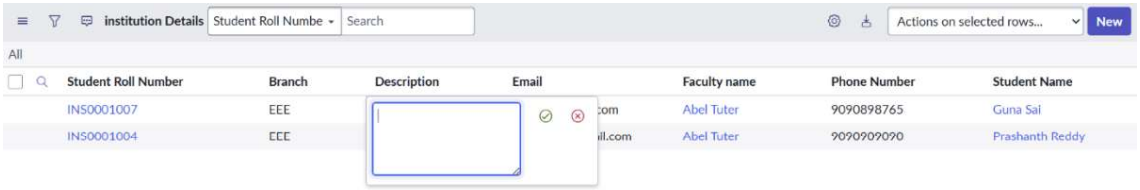
- Click New to create a new ACL(Write)
- Enter the following details
- Type: record
- Operation: Write
- Name: u_institution_details
- Active: true
- In the Requires role related list, add: bb3 (custom role)
- No need to add any Data condition
- Save the ACL Record

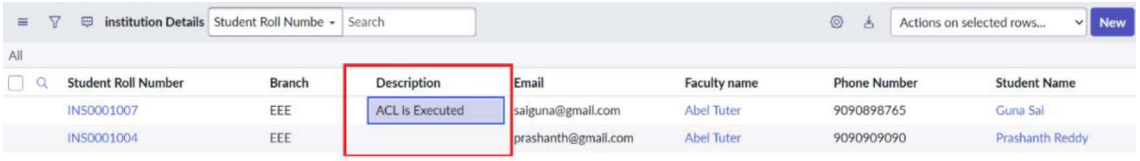




Verify :

- User with **bb1 &bb2 &bb3 roles** can view and edit the **EEE branch records along with New** Button

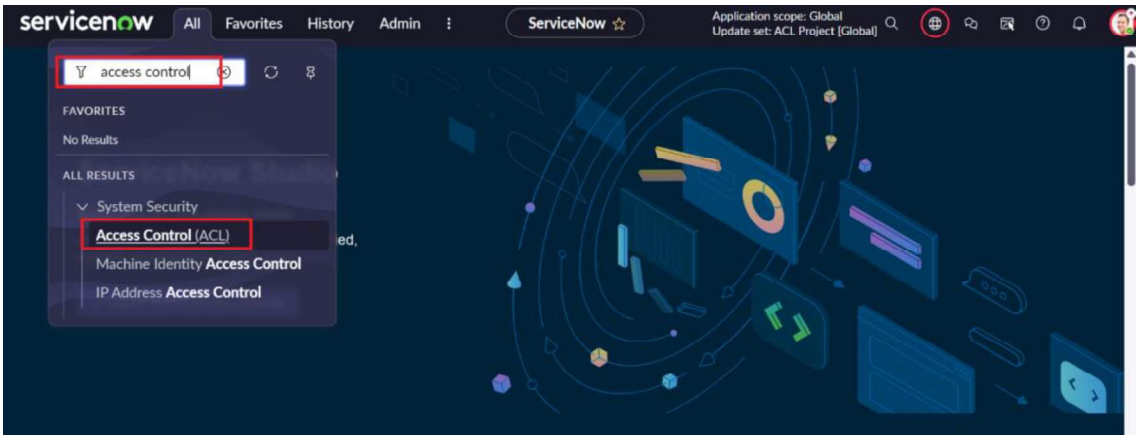


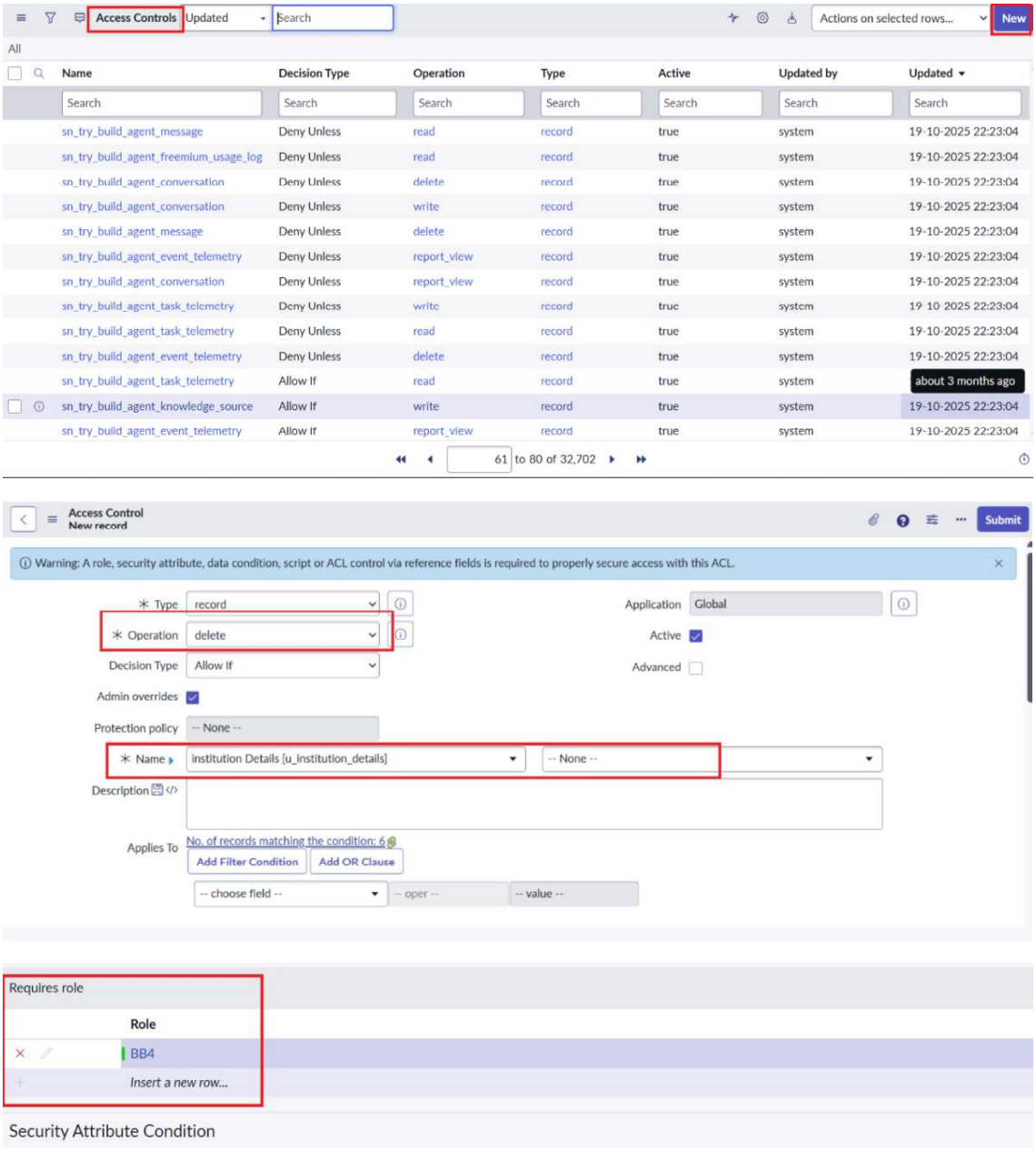


Creation of ACL-DELETE:

Description:

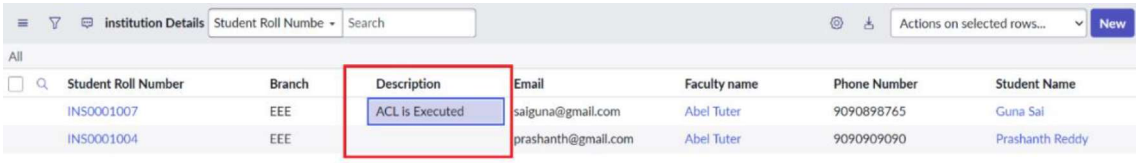
- Click New to create a new ACL(Delete)
- Enter the following details
- Type: record
- Operation: Delete
- Name: u_institution_details
- Active: true
- In the Requires role related list, add: bb4 (custom role)
- No need to add any Data condition
- Save the ACL Record

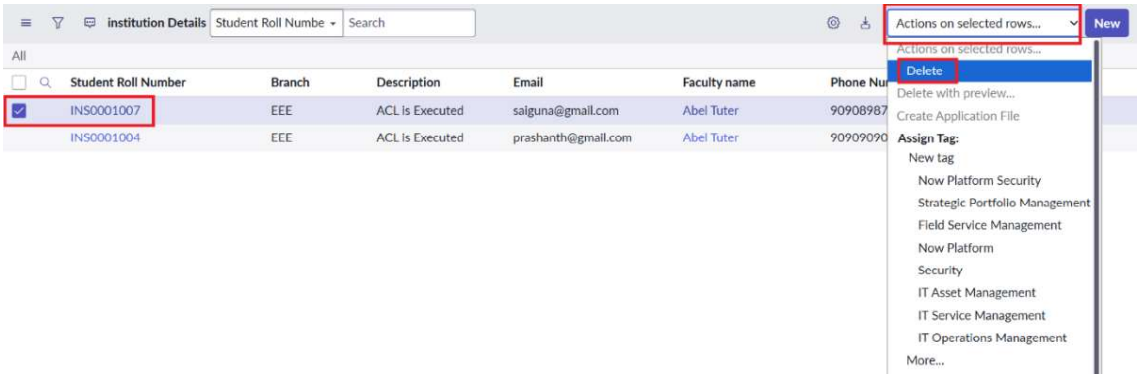


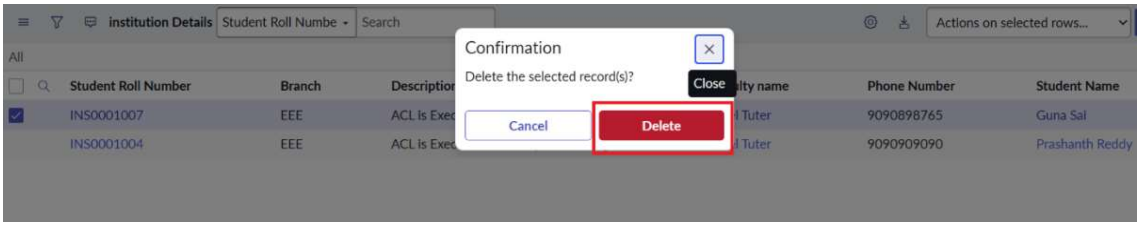


Verify :

- Users with **bb1 &bb2 &bb3&bb4 roles** can see, create and edit along with delete the **EEE branch records.**







Conclusion:

Description:

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.